CHAPTER 6

The structure and functioning of African swamps

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In Africa the term 'swamp' generally applies to wet areas covered with tall herbaceous vegetation. Beadle (1974) calls such areas 'shaded swamps'. Swamps are basically aquatic systems, and the food web is largely aquatic, although limited interaction with surrounding terrestrial areas may occur. Therefore, the ecology of swamps is different from floodplains and other moist habitats which are only temporarily covered with water. Swamps are characterised by the presence of relatively tall vegetation. This tall vegetation has a significant effect on the ecosystem which is not found in other moist areas such as bogs, where the vegetation cover is short. In Africa herbaceous swamps can be defined as: flat areas which are flooded to a shallow depth either permanently, or for most of the year, and which are densely covered with herbaceous vegetation whose shoots rise out of the water to a height of more than one metre. Such swamps can be bottom-rooted or floating (Fig. 6.1).

In temperate regions swamp plants are normally bottom-rooted, good examples being the *Phragmites* grass marshes of Middle Atlantic USA or Eastern Europe, the *Typha* cattail or sedge marshes of Europe and Mid-western USA, the sedge marshes (*Cladium*) of Great Britain, the Everglades, *etc.* Tropical *Phragmites* and *Typha* marshes are structurally similar to their temperate counterparts. Floating swamps are more common in the tropics than elsewhere – for example, the floating grass meadows of South-east Asia and South America, or the *Vossia* swamps along African rivers (Sculthorpe, 1967; Junk, 1970). In Africa, floating *Vossia* or papyrus swamps (*Cyperus papyrus* L.) are structurally quite different from bottom-rooted swamps.

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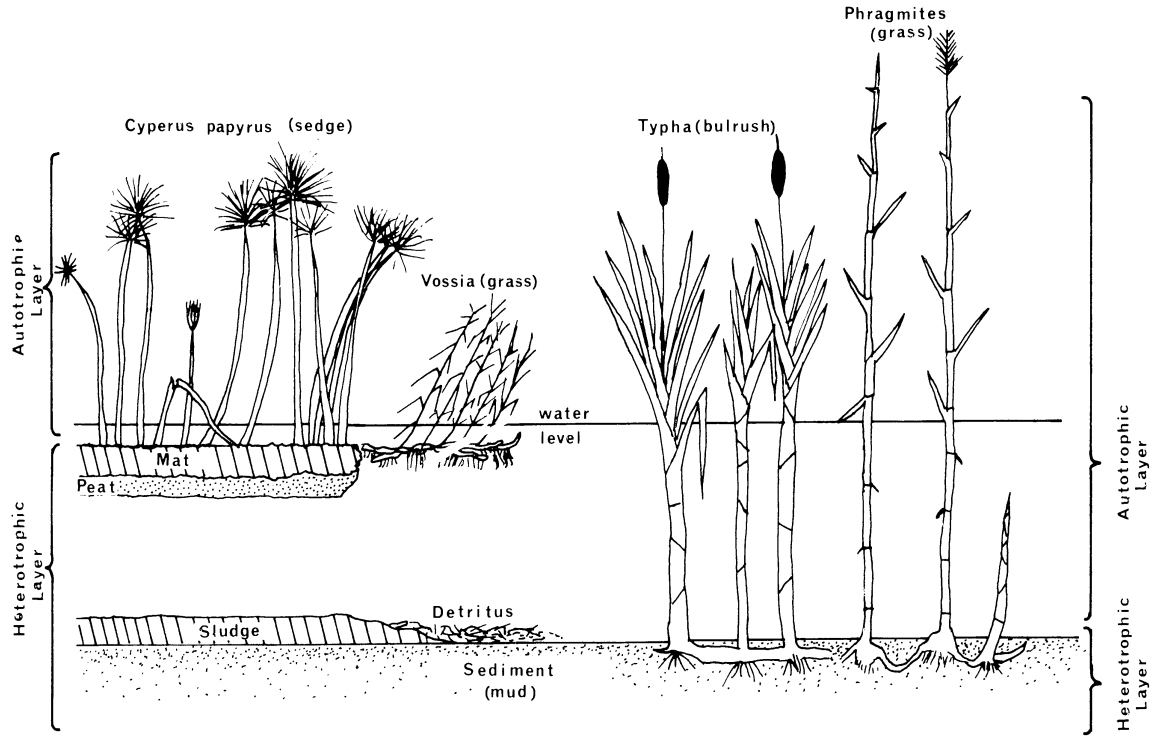

Fig. 6.1. Comparison of floating and rooted tropical swamps showing differences in general structure.

6.1 Basic structure of herbaceous swamps

Swamps occupy an ecological position between purely aquatic and purely terrestrial environments, and thus the structure of swamp ecosystems is rather different from both. Odum (1971) considers that ecosystems can be separated vertically into two layers, the autotrophic (producer) and the heterotrophic (consumer) layers. In swamps the bulk of the primary production occurs in air, and consumption and decomposition in water, thus producer and consumer organisms are physically separated (Fig. 6.1). This spatial separation between autotrophic and heterotrophic components leads in turn to a separation in time, because the foodweb has no direct access to the freshly produced material. There is, therefore, a timelag between production and the death and collapse of the produced material into the water where it can be attacked by aquatic organims. Such a space-time separation between ecosystem components is typical of systems in which the food web is based on detritus (Teal, 1962; Odum, 1971). Whilst this generalisation on trophic stratification is applicable to tropical swamps, this is not the case with modes of nutrient cycling, where we can recognise two very different systems reflected in the gross physical structure of two types of African swamp. The autotrophic component can either be rooted in the sediment (such as in swamps dominated by *Typha*, *Phragmites*, *Cladium* etc.) or rooted in a floating or almost floating mat (such as in swamps dominated by *Cyperus papyrus*, *Vossia* etc.). In the first type, the sediments can act as a source of nutrients for the overlying water column, whilst in the second type the sediments or bottom deposits act as a sink for nutrients from the water column. The two types of swamp (Fig. 6.1) are referred to here as *rooted swamps* and *floating swamps* respectively. Apart from showing different nutrient pathways, as will be discussed later, these two swamp types have a number of different environmental characteristics.